

HARVARD UNIVERSITY

FACULTY OF ARTS AND SCIENCES

SUMMER SCHOOL
Pre-College Program



51 Brattle Street
Cambridge, Massachusetts
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Sewon Lee

Introduction to Data Science with a Focus on Visualization CSCI P-14115 (35871)

July 8 - July 19, 2024

This course is a short introduction into the world of data science with a heavy focus on visualization. The course is split into two parts. In the first week, students learn to use Python to compile clean and convenient-to-use data sets from the web and explore this data through visualizations using the Python library Matplotlib. This first phase concludes with a two-day in-class guided mini project, in which students use their newly acquired knowledge to create visualizations based on COVID-19 infection data. In the second week, we take an in-depth look at visualizations and their communicative potential. Together, we create more complex and dynamic visualizations using JavaScript. Throughout this course, we guide students through the computational analysis of different data sets, starting with web scraping and data cleaning and ending with implementing sophisticated visualizations to present findings. We do not lecture about computational data analysis or computer programming; rather, students learn about these things deeply, collaboratively, and in an environment where everyone is dependent upon each other's contributions.

Sewon performed excellently in CSCI P-14115 and was among the top students. He submitted all work on time and participated actively in discussions. Sewon's assignments were always of excellent quality. He produced outstanding work, was very skilled in producing complex programs, and showed a great talent for quickly understanding and effectively applying abstract computer science concepts. To conclude, Sewon is a strong and diligent student. He was genuinely interested in the course material and went beyond what was expected.

A handwritten signature in black ink, appearing to read 'R. Roessler'. The signature is fluid and cursive, with a long horizontal stroke at the end.

Robert Roessler
Resident Tutor, Harvard University